Fiscal Year:	FY 2006	**Task Last Updated:**	FY 09/13/2005
PI Name:	Pishko, Michael V Ph.D.		
Project Title:	Microfabricated Optical Biosensor Arrays for Air Quality Monitoring		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced environmental monitoring and control		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	16802	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-01
Start Date:	10/01/2001	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	3
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG9-1372		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	We propose to design and prototype biosensor array systems capable of optically detecting and identifying a wide variety of airborne trace organic, potentially toxic agents. These compounds include ketones, aldehydes, alcohols, alkanes, and halocarbons. This approach to the problem combines basic biological and materials science with novel microfabrication techniques. In specific we will immobilize fluorophore-labeled biorecognition molecules (apoenzymes) within novel poly(ethylene glycol) based gel materials that maintain protein conformation and thus biological function, and that can be patterned via photolithography or microcontact printing. These gels will be designed to allow partition of the agent from the gas phase into the gel where it can bind to the recognition molecule. To achieve optical detection, a fluorophore-modified ligand analogous to the agent of interest will be immobilized in the gel. In the absence of the agent, the ligand and recognition molecule will bind, resulting in fluorescence resonance energy transfer (FRET) between the respective fluorophores. When the agent is present, the ligand will be displaced competitively, FRET will cease, and the fluorescence in the gel increase. By micropatterning an array of these gels with different recognition		

	molecules, multiple analytes can be simultaneously detected either directly or via pattern recognition techniques. This project will focus on the development of a fluorescent microsensor array for the following six model analytes: methanol, isopropanol, acetone, formaldehyde, chloroform, and toluene. Once patterned into an array via photolithography, the change in FRET upon exposure to gaseous analytes will be monitored using a CCD camera. The creation of such an arrays would represent a unique, specific, sensitive biologically-inspired method of gaseous agent detection in manned vehicles. Develop optical sensor arrays for volatile organic compounds, using biological recognition molecules to achiece recognition.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The identification and measurement of toxic volatile organic compounds and airborne pathogens are of critical concern not only to manned space flight, but to all Americans. There are numerous examples where sensors for these materials would be extremely useful. The bioterrorism attacks of 2001 were the result of exposure to an airborne pathogen, anthrax. Many rescue workers at the World Trade Center site suffer from respiratory problems, the result of prolonged exposure to toxic volatile organic compounds and smoke produced by the burning debris. Accounts of office workers becoming ill from toxins and pathogens in building ventilation systems ("sick building syndrome") have frequently occurred over the last two decades. Unfortunately, in all these cases, there are no rapid and accurate methods for quantifying airborne toxins and pathogens. This project seeks to create fast-response multianalyte sensors for these materials using biological inspired methods.
Task Progress:	Gel microstructures based upon poly(ethylene glycol) diacrylates, dimethacrylates, and tetraacrylates patterned photolithographically on silicon or glass substrates were created and characterized. A silicon/silicon dioxide surface was treated with 3-(trichlorosilyl)propyl methacrylate to form a self assembled monolayer (SAM) with pendant acrylate groups. SAM presence on the surface was verified using ellipsometry and time-of-flight secondary ion mass spectrometry. A solution containing an acrylated or methacrylated poly(ethylene glycol) derivative and a photoinitiator (2,2-dimethoxy-2-phenyl-acetophenone) was spin-coated onto the treated substrate, exposed to 365 nm ultraviolet light through a photomask and developed with either toluene, water, or supercritical CO2. As a result of this process, three-dimensional, cross-linked PEG gel microstructures were immobilized on the surface. Diameters of cylindrical array members were varied from 600 um to 7 um by the use of different photomasks, while height varied from 3 to 12 um, depending on the molecular weight of the PEG macromer. In the case of 7 um diameter elements as many as 400 elements were reproducibly generated in a 1 mm2 square pattern. The resultant gel patterns were solvent-swollen for as long as three weeks without delamination from the substrate. In addition, micropatterning of different molecular weights of PEG was demonstrated. PEG gel microstructures were also created and arrayed using a microspotting robot, similar to that used to created DNA microarrays. Using this micropatterning technology, arrays of gels capable of detecting multiple analytes may be created without repeated lithographic patterning procedures. We demonstrated that these gel microstructures could be integrated with microfluidic systems for sample deliver via either a gas or liquid stream.
Bibliography Type:	Description: (Last Updated: 08/24/2007)
Articles in Peer-reviewed Journals	Koh W-G, Pishko M. "Immobilization of multi-enzyme microreactors inside microfluidic devices." Sensors and Actuators B: Chemical, 2005 Apr;106(1):335-42. http://dx.doi.org/10.1016/j.snb.2004.08.025 , Apr-2005
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Articles in Peer-reviewed Journals	Yadavalli VK, Koh WG, Lazur GJ, Pishko MV. "Microfabricated protein-containing poly(ethylene glycol) hydrogel arrays for biosensing." Sensors and Actuators B: Chemical, 2004 Feb;97(2-3):290-7. http://dx.doi.org/10.1016/j.snb.2003.08.030 , Feb-2004